

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester B. Tech. (ME) Examination****April-May 2018****ME 245 Manufacturing Processes - II****Date: 30.04.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.**

- (i) Give specification criteria for Capstan and Turret Lathe. **[03]**
- (ii) A mild steel rod having 50 mm diameter and 500 mm length is to be turned on a lathe. The rod is to be reduced to 45 mm in one pass when cutting speed is 30 m/min and a feed of 0.7 mm/rev is used, find: **[04]**
- i. Cutting time.
 - ii. Material removal rate.

Q - 2. Answer any three questions.**[12]**

- (i) List the operations performed on Lathe machine and explain any three with neat sketches.
- (ii) Explain following with neat sketch: **1.** Lathe centers. **2.** Rests.
- (iii) What do you understand by the term “boring”? Why it is required?
- (iv) List various clamping devices used for holding the work on drilling machine and explain any two with neat sketch.
- (v) Calculate the time required to drill a hole 30 mm diameter in a cast iron workpiece 80 mm thick using a high speed steel drill. The cutting speed and the feed rate for the operation may be assumed to be 25 m/min and 0.3 mm per revolution of the drill respectively.

Q - 3 Answer any four questions.**[16]**

- (i) A 200 mm long and 100 mm wide surface of a cast iron workpiece is to be rough machined using 150 mm diameter face milling cutter with 16 teeth. The speed, feed and depth of cut for operation are 0.33 m/sec, 0.05 mm/tooth and 6 mm respectively. Find machining time and material removal rate.
- (ii) Sketch & compare up milling and down milling.
- (iii) Differentiate between Shaper and Planner with reference to its construction, operation and applications.
- (iv) Explain the grinding wheel – workpiece interaction.
- (v) Explain the following operations:
 - i. Through feed grinding
 - ii. In feed grinding
- (vi) State advantages and limitation of broaching.

SECTION – II

Q - 4 Answer the following.

- (i) Justify the statement: A hook made by forging process will have better mechanical properties rather than produced by machining processes. [02]
- (ii) Residual stresses play a major role in manufacturing industry. Discuss. [04]
- (iii) Differentiate hot metal working process and cold metal working process. [05]

Q - 5 Answer any three questions.

[12]

- (i) Explain steam type hammer with neat sketch.
- (ii) Explain backward extrusion process with neat sketch.
- (iii) Write a short note on any two types of rolling mill with neat sketch.
- (iv) In Tata steel industry cold rolling process is done on a 300 mm thick slab when $\mu = 0.08$ and the roll radius is 300 mm. Determine maximum possible reduction. On the same mill find out maximum reduction during hot rolling when $\mu = 0.5$?
- (v) Determine the engineering strain and true strain for (1) a bar which is doubled in length (2) a bar which is halved in length.

Q - 6 Answer any three questions.

[12]

- (i) Differentiate open die forging and close die forging process.
- (ii) What do you mean by extrusion ratio? Shree ram extrusion Pvt. Ltd. company makes a bar of 75 mm^2 from a billet having cross section area 3600 mm^2 . Find the extrusion ratio for it.
- (iii) How the tube drawing process is done by using mandrel? Draw the neat sketch.
- (iv) How the punching operation differs from blanking operation?
- (v) "C" shaped parts are produced from 3 mm thick C20 steel sheet as shown in figure 1. Find the length of sheet required for it. Both bend angles are 90° .

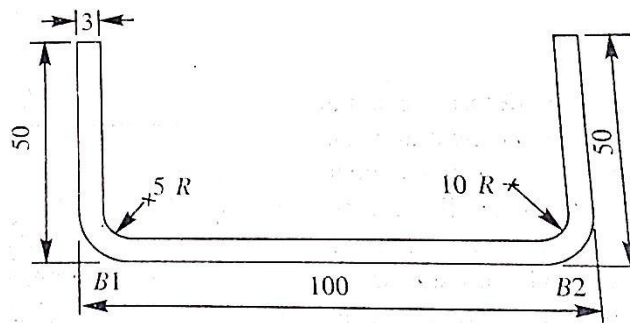


Figure 1
